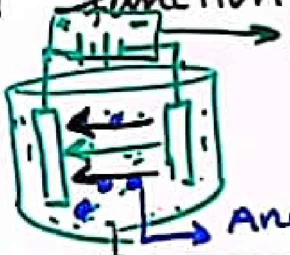
Voltametry:-

Volt → measurement

Definition:-

Voltametry is a electrochemical technique in which current at electrode, is measured. as a function voltage or potential difference.

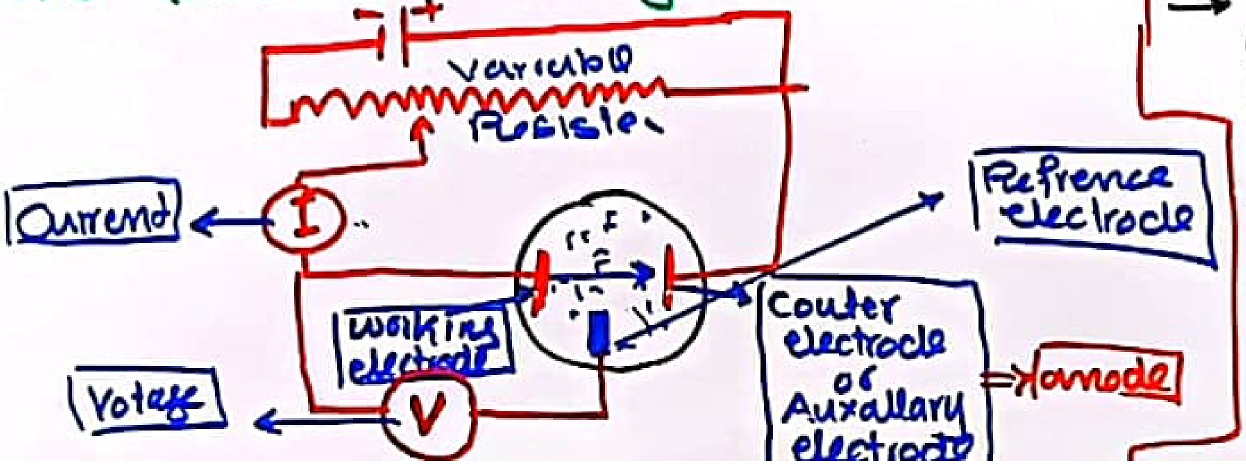


Analyte → Ions → Cations (+ve) / Anions (-ve) — electro active specie.

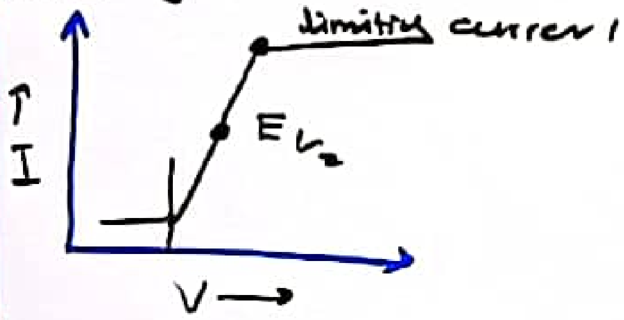
$\boxed{I \propto \text{Concentration}}$

20 → 0.5A
100 → 2A

Purpose:- This technique is used for qualitative and quantitative analysis.



- → It is consist of
- → Working electrode, at this electrode oxidation or reduction occur.
- → Refrence electrode having fix potential difference. (SCE).
- → Couter electrode or Auxallary electrode.
- → A battery
- → A variable resister used to vary the voltage.
- → Voltmetermeter, which is connected b/w working electrode and refrence electrode.
- → Ammeter connet the working electrode and auxallary electrode.

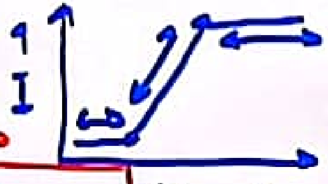


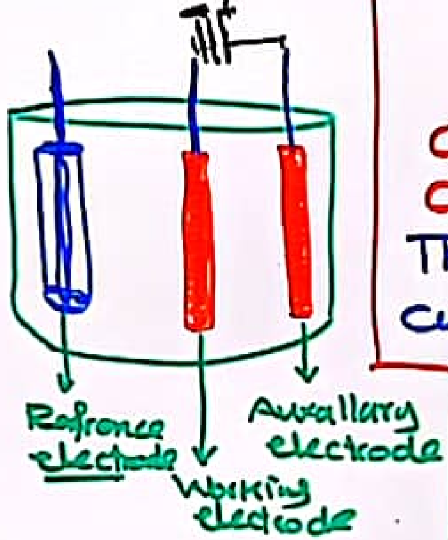
Voltammetry:-

"Votammetry is the electrochemical technique in which current at an electrode is measured as a function of the voltage"

Votamogram:-

The graph which is obtain after voltametry, b/w voltage and current is called .....





Current

Faradaic Current

The current which is produced due to oxidation (lose of e⁻) or reduction (gain of electron) is called faradic current

Non Faradaic current

The current which is produce not for oxidation or reduction is called .....

Faradaic Current

Cathodic Current

The faradaic current which is produced due to reduction of analyte is called cathodic current.

$Cd^{+2} + 2e^- \rightarrow Cd$

$Cu^{+2} + 2e^- \rightarrow Cu$

Anodic current -

The faradaic current which is produced due to oxidation of analyte is called anodic current.

$2Cl^- \rightarrow Cl_2 + 2e^-$

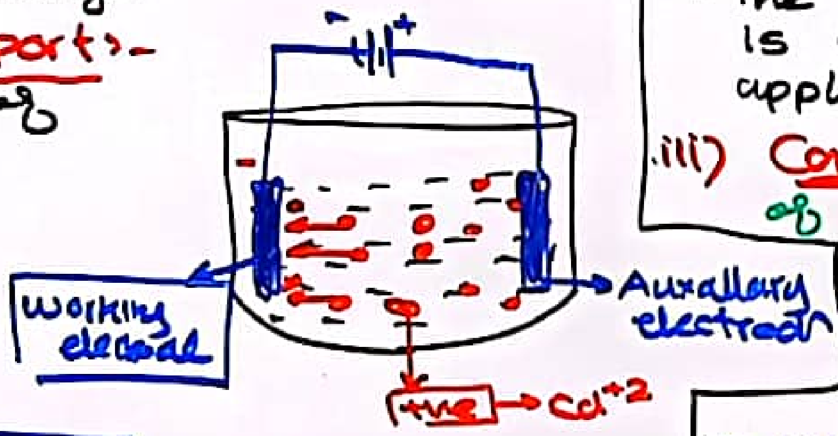
Voltammetry :- (Polarography).

"Votammetry is the electrochemical technique in which current at an electrode is measured as a function of the voltage"

Modes of Mass transport :-

There are three modes of transport in voltametry.

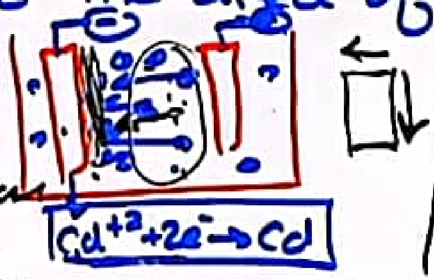
i) Diffusion.
ii) Migration.
iii) Convection.



Diffusion :-

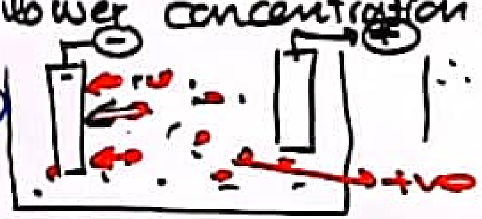
The movement of analyte ions from the area of higher concentration to the area of lower concentration....

→ when electric current is passed in the electrode the the analyte ions near the electrode oxidize or reduce. and concentration of ions become smaller near the electrode and ions move from the area of higher to the area of lower concentration.



Migration :-

The movement of cation or anion toward electrode when voltage is applied is called migration.



→ Anion move toward anode while cations move toward cathode.

→ The movement of ions is depend on the applied voltage.

iii) Convection :-

The movement of material in response to a mechanical force, such as stirring a solution.

Only Diffusion current is more important and other currents are eliminate.

$$i = \frac{nFAD\,(C_{bulk} - C_{x=0})}{\delta}$$

∴ n = no. of electron transfered in a redox reaction.

∴ F = Faraday constant

∴ A = Area of electrodes.

∴ D = The diffusion co-efficient for reactant and product

∴ C_{bulk} and $C_{x=0}$ = The concentration of analyte in bulk solution and at the electrode surface

∴ δ = Thickness of diffusion layer

Voltammetry:-

"Votammetry is the electrochemical technique in which current at an electrode is measured as a function of the voltage"

Instrumentation of Voltametry:-

There are basically three component of Voltametry.

1. Electrode
2. Potentiometer.
3. Voltametric cell solution.

Electrodes:- There are three types of electrodes.

1. Working electrode
2. Auxallary electrode
3. Refrence electrode.

Working electrode:- The electrode where oxidation or reduction of analyte occur is called.

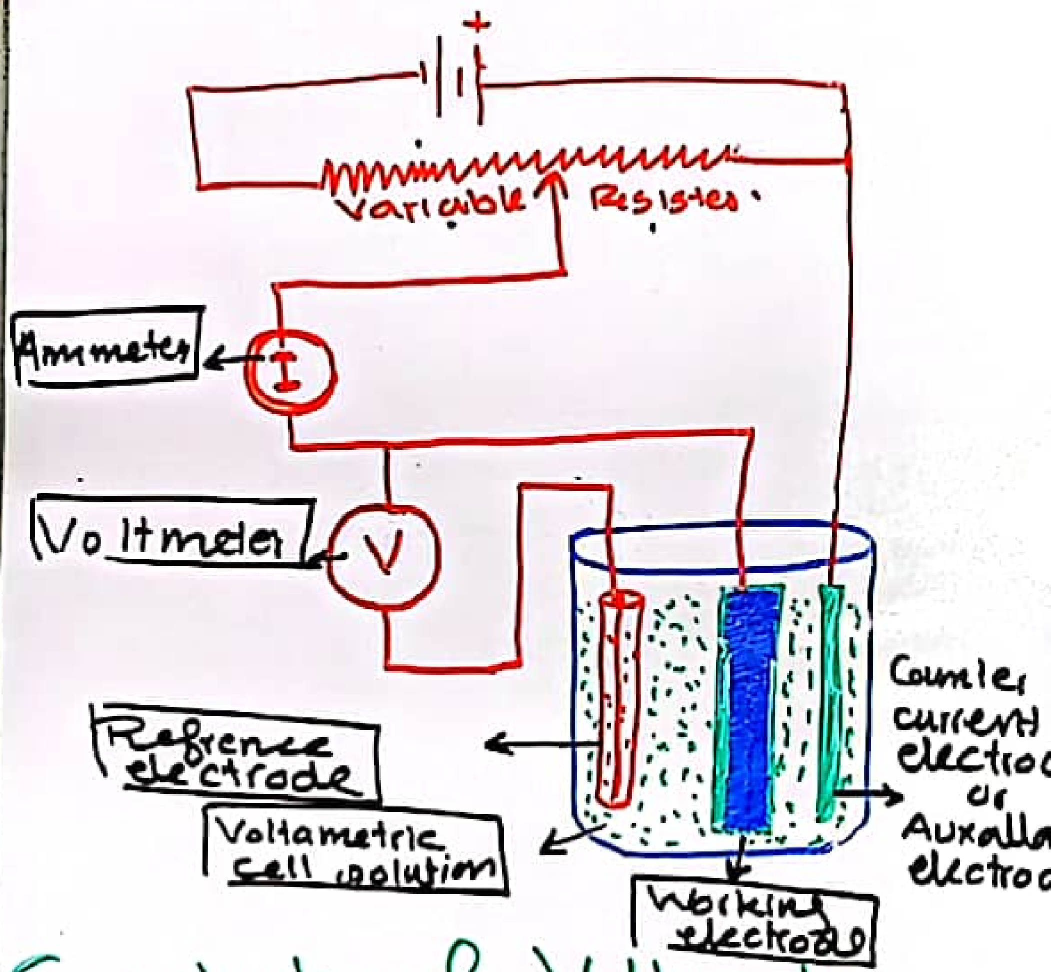
→ Potential difference is continuously charge b/w working electrode and refrence electrode.

→ Voltmeter is placed b/w working electrode and refrence electrode.

Examples:-

→ Hanging drop mercury electrode (HDME) (used in the ppb to ppm range)

→ Static drop mercury electrode (SDME) (used in low ppm range).



Construction of Voltametry.

Voltammetry:-

Votammetry is the "electrochemical technique in which current at an electrode is measured as a function of the voltage"

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→ Voltmeter is placed b/w working electrode and refrence electrode.

Examples:-

→ Hanging drop mercury electrode (HDME) (used in the ppb to ppm range)

→ Static drop mercury electrode (SDME) (used in low ppm range).

(Ppm range)

• Rotating Disk electrode (RDE)
→ Ultra trace graphite. → Gold.
→ Glassy carbon.

Auxallary electrode:-

→ Auxallary electrode is used to complete the circuit.

→ The current is flow b/w working electrode and Auxallary electrode.

→ Material: Pt, Glassy Carbon.

Refrence electrode:-

→ The potential difference of refrence electrode is fix.

→ By using refrence electrode we determine the potential difference of working electrode.

→ Voltmeter is present WE & RE

Examples:-

→ Calomel electrode

→ Ag/AgCl electrode.

Voltammetry:-

Votammetry is the "electrochemical technique in which current at an electrode is measured as a function of the voltage".

Instrumentation of Voltametry:-

There are basically three component of voltametry.

1- Electrode
2- Potentiostat.
3- Voltametric cell solution.

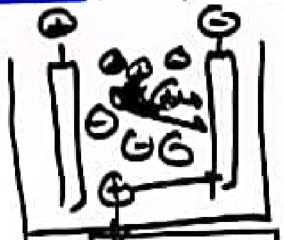
Voltametric cell solution:-

Voltametric cell ~~or~~ solution is consist of two parts.

Supporting electrolyte — Analyte

Analyte → Ions

Cations | Anions



Migration

1 : 100

Supporting electrolyte:-

supporting electrolyte is a salt which

→ reduce the effect of migration.

→ lower the resistance of solution.

Characteristics of supporting electrolyte.

→ should be chemically inert

→ do not interfere with diffusion and the electrons exchange on the electrode surface.

→ have different discharge potential. (at least 100 - 200 mV)

0.4 - 0.8 mV

→ have an high ionic conductivity.

→ Adjust the pH

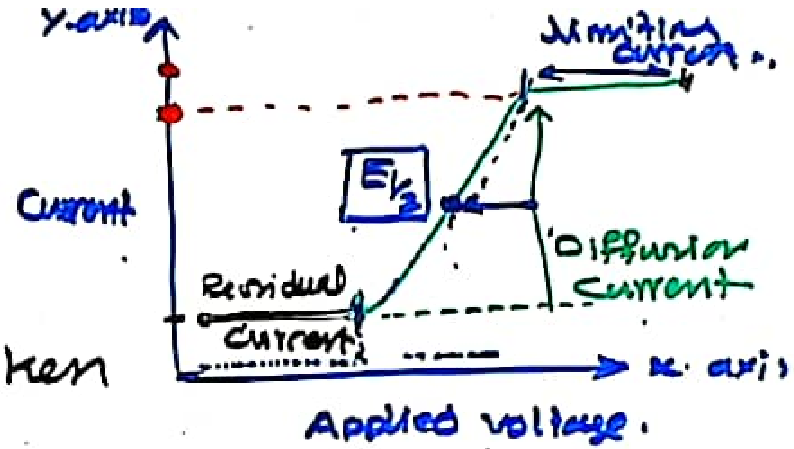
→ increase the selective.

Voltametry:-

Voltamogram:-

→ 50 mV/second

After performing voltametric technique a graph is obtained b/w voltage and current is called ....



Voltamogram is formed b/w voltage and current.
→ Voltage is taken on x-axis and current is taken on Y-axis.
→ A S-shap graph is obtained.

Residual current:-
→ A small current is obtain even when no analyte is present inside the solution is called ......
→ Residual current may be due to following reasons.

✓ The residual current may be due to some impurities and the current may be faradic or non-Faradaic

✓ A charging current because of electrically double layer.

Diffusion Current:- Diffusion current is obtained due to the oxidation or reduction of analyte ion.
→ when we increase the voltage the oxidation or reduction of analyte is also increase with the increase of oxidation or reduction the value of diffusion current is

→ Half wave potential is used for qualitative analysis. Each ion has specific half wave potential.

Limiting Current:-
Limiting current is depend upon the concentration of analyte.

$$i_L = K C_A.$$

where
i_L = Limiting Current.
C_A = Concentration of analyte.
K = Constant.

Limiting current is directly proportional to the

Voltammetry:-

Different technique of Voltammetry:-

1- Polarography
2- Hydrodynamic Voltammetry.
3- Stripping Voltametry.

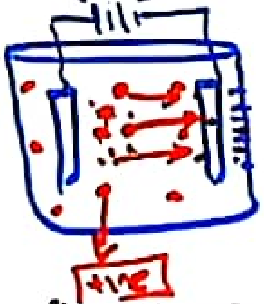
Hydrodynamic Voltametry:-

water movement.

Hydrodynamic voltametry is a type of Voltametry in which the analyte solution is kept in continuous motion.

Mode of mass transfer:-

All the three modes of mass transfer is worked.

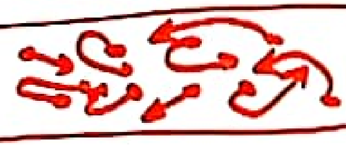


→ migration under the influence of electric field.
→ Convection resulting from stirring.
→ Diffusion due to concentration difference b/w the electrode surface and bulk of solution.

Flow of Liquid:-

The flow of Liquid may be two types.

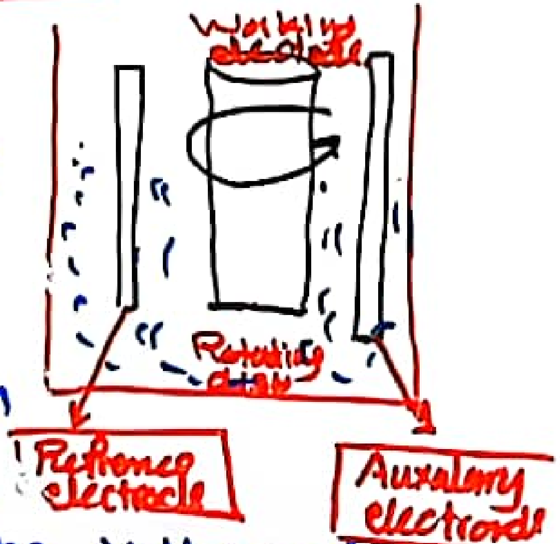
Turbulent flow:- Turbulent flow happen at high velocities and has irregular fluctuation motion.



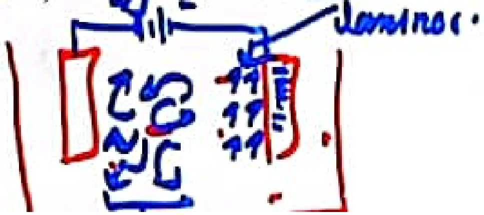
• **Laminar Flow:-** Laminar flow occur at low velocity. The molecules has smooth and and regular motion.

Construction & working:

→ It is also consist of three electrode
• Refrence electrode
• Auxallary electrode
• working elector or Rotatory disk.

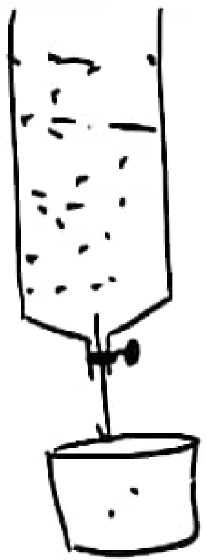


→ Analyte solution is added and stirr the solution and note down the Voltage change with current and Voltamogram is obtained.



Current product is depmoon
Convection → electrode surface

Applications of Hydrodynamic Voltametry:-



i) Detection and determination of chemical specie as the exit from chromatographic colums or continuous flow apparatus.

ii) Routene determination of oxygen and certein specie of biological interest. such as glucose lactose & sucrose etc.

iii) It is used for the detection of end point in column chromatography.

iv) It is used for fundamental studies in electro chemical process.

v) It is used for qualitative & Quantitative analysis.

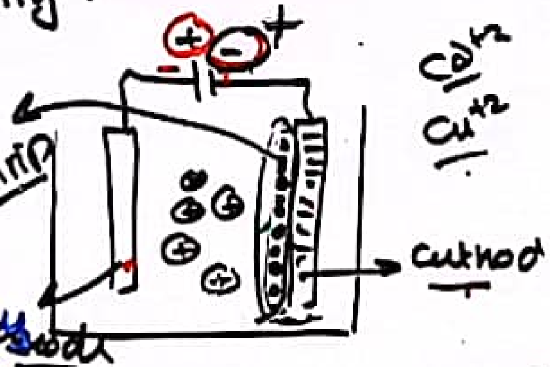
vi) This technique is also used for the determination of purity of a substance.

Different Voltametric Technique:-

→ Hydrodynamic Voltametry.
→ Stripping Voltametry.
→ Polarography
→ cyclic volltametry.

Stripping Voltametry:- strip

Take off. / lift



Definition:-

Stripping Voltametry consist of applying two opposite potential volume where oxidation take place at one value and reduction take place at another value.

+1.0, 0.9, 0.8, 0.7 ..... 0 ..... -0.1, -0.2 ....

→ In stripping voltametry, the analyte is first deposit and then removed or stripped electrochemically while moniting the current as the function of applied voltage.

Steps:- It is consist of two steps.

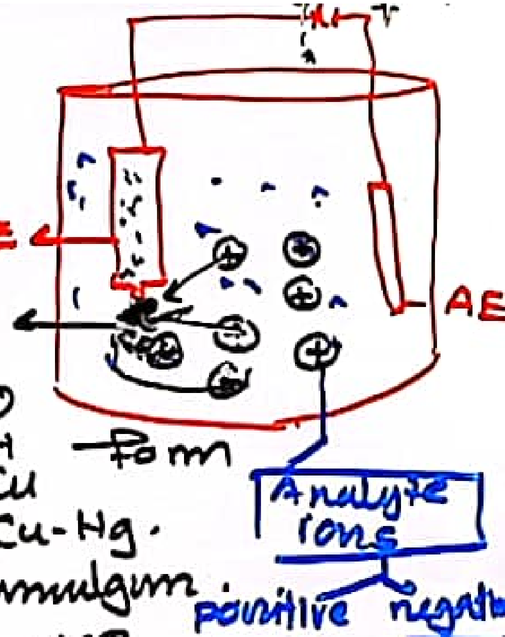
i) Pre-concentration Steps:-

→ In stripping Voltammetry hanging mercury electrode is used

→ electrodes of Noble metal also used in this voltametry. (Pt, Au)

→ The analyte ions may be +ve or -ve

→ When cation is present in the solution it attract toward Hg electrode and reduce. and it form



$Cu^{+2} + 2e^- \rightarrow Cu$

$Cu + Hg \rightarrow Cu\text{-}Hg$.

Mercury metal amulgum.

→ The anion move toward electrode (Hg) and form salt. $Cl^{-1} \rightarrow Cl_2 + 2e^-$

$Hg + Cl_2 \rightarrow HgCl_2$.

ii) Stripping step:-

Now potential difference is vary(change) in such a way that the analyte ions stripp from electrode.

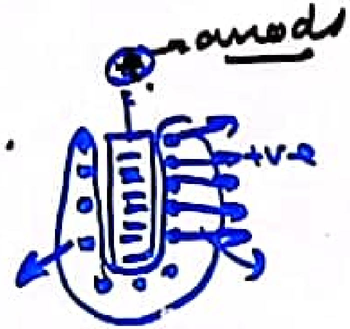
From whole process the voltage & current is measured.

Different Voltametric Technique:-

→ Hydrodynamic Voltametry.
→ Stripping Voltametry.
→ Polarography
→ cyclic volltametry.

Stripping Voltametry:-

There are three types of stripping.
→ Anodic stripping voltametry
→ Cathodic " "
→ Adsorptive " "



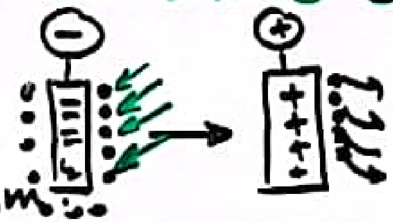
Anodic stripping voltametry:-

If stripping give anodic current is called ----------

→ At cathode any cation attached.

$Cu^{+2} + 2e^- \rightarrow Cu$

$Cu + Hg \rightarrow Cu(Hg)$ amalgm



The anodic stripping occur when scan potential from positive to negative.

$Cu(Hg) \rightarrow \underbrace{Cu^{2+} + Hg + 2e^-}_{\text{oxidation} \rightarrow \text{anode}}$ anodic current

→ Anodic striping Voltametry is applicable for those ions who form amalgm with mercury e.g Cd^{+2}, Cu^{2+}, Bi^{3+}, Pb^{2+}, Ni^{+2} etc.

Cathodic stripping Voltametry:-

If stripping give cathodic current is called ------

$Hg \rightarrow Hg^{+1} + 1e^-$

$2Hg^{+1} + X_2 \rightarrow$

Hg_2X_2. insoluble

→ During stripping.

$\underbrace{Hg_2X_2 + 2e^-}_{\text{Reduction} \downarrow \text{Cathode}} \rightarrow 2Hg + 2X^-$

Applications of Stripping Voltametry:-

→ It is used for qualitative analysis & Quantitative analysis.

→ This technique is used for the analysis of Cu, Zn & Mn in soil & vegetable sample.

→ It is used for the determination of concentration of Pb^{+2} ions in water.

→ This technique is used for the determination of.

- ✓ food Contimination
- ✓ Toxic metals
- ✓ Pesticide
- ✓ fertilizers.
- ✓ Trace essential element
- ✓ food additive dye.
- ✓ organic compounds of biological significance.

Cyclic Voltametry:- $\boxed{-0.4} \to 0 \to \boxed{0.4}$

0

cycle.

cyclic voltametry is an electrochemical technique. for the measuring the current of a redox active analyte. by scanning the potential b/w two or more set value in both forward and backward direction. ..... –

→ It is used for qualitative analysis and quantitative analysis.

→ After cyclic voltametry a graph is obtain having "duck shape"

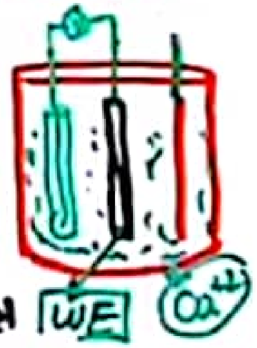
→ In cyclic voltametry three electrodes are used.

i) Working electrode ii) Refrence electrode

iii) Auxillary electrode / counter electrode

Graph:- when we increase the voltage from –ve potential to +ve potential. oxidation of analyte occur

$$R \longrightarrow O + ne^-$$



At one point maximum current is obtain which is called anodic peak current. (i_{pa})

→ The potential difference where maximum current is obtain is called anodic peak voltage.

At one point the potential is reversed then the potential difference where current reverse is called swtiching potential.

when we move backward (from +ve to –ve potential) then reduction of analyte occur. .

$$O + ne^- \longrightarrow P.$$

→ In reverse maximum current is obtain a specific voltage is called cathodic peak current.

→ The potial where i_{pc} is obtain is called E_{pc} (cathodic peak voltage.

Cyclic Voltametry potential wave form:-

Cycllc Voltametry:-

An equation is used to determine the peak current.

$$i_p = (2.69 \times 10^5)\, n^{3/2}\, A C D^{1/2}\, v^{1/2} = KC$$

The above equation is called Randles-Sevcik equation.

where.

∴ i_p = Peak current.

∴ n = no. of electron in redox reaction.

∴ A = Area of electrode.

∴ C = concentration of solution.

∴ D = Diffusion coefficient for electroactive specie.

∴ v = Scan rate.

→ The potential difference where maximum current is obtain is called anodic peak voltage.

At one point the potential is reversed then the potential difference where current reverse is called switching potential.

when we move backward (from +ve to -ve potential) then reduction of analyte occur.

$$O + ne^- \longrightarrow R.$$

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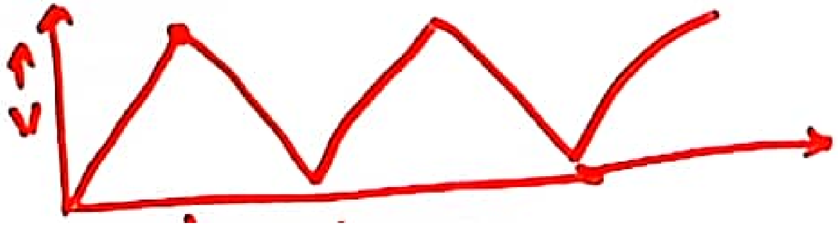

Cycle Voltametry:- $\boxed{-0.4} \rightarrow$ c

cycle.

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used.

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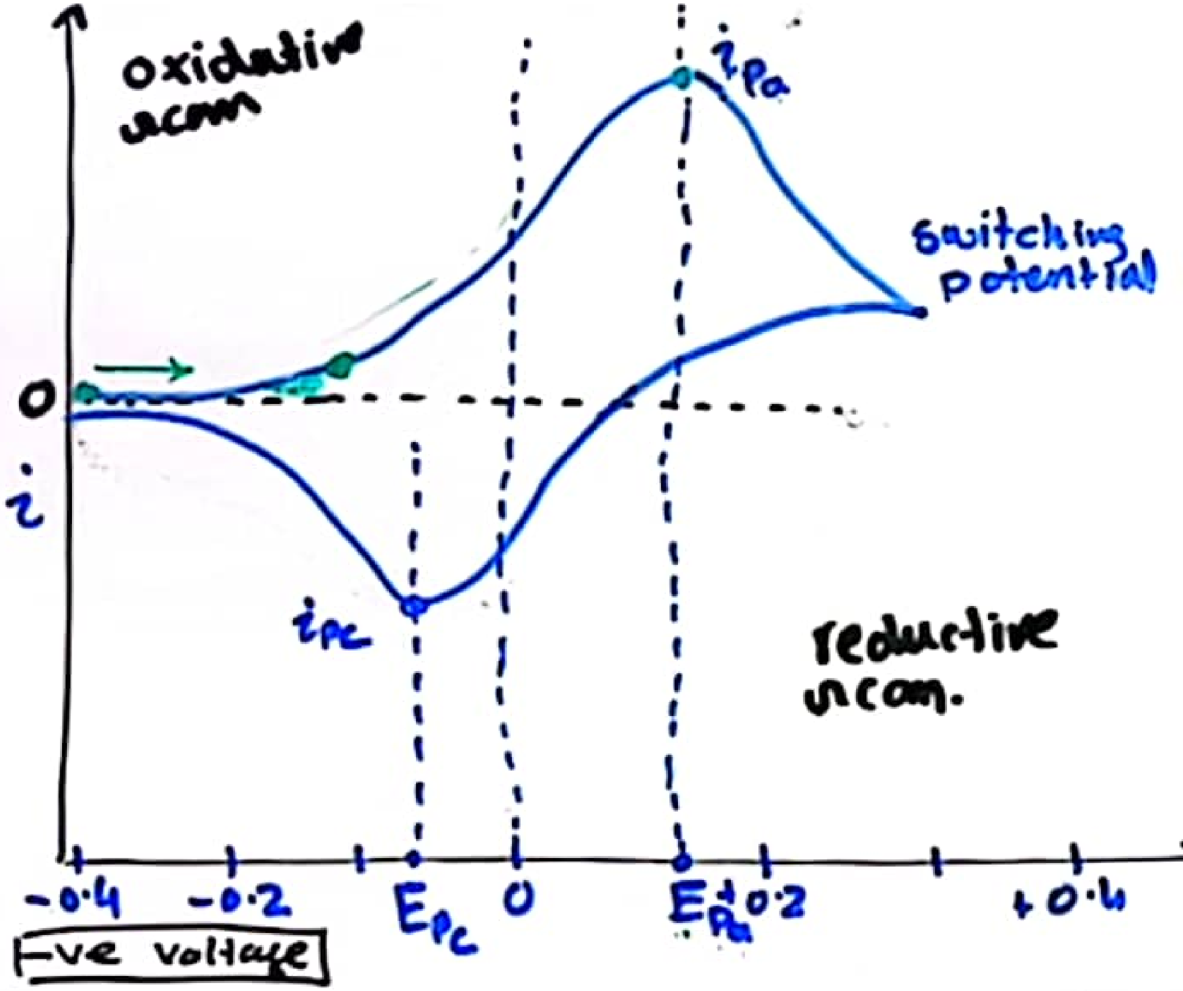
Graph:- when we increase the voltage from -ve potential to +ve potential. oxidation of analyte occur



Cyclic Voltammetry:-

An equation is used ... the peak current.

$i_p = (2.69 \times 10^8)\, n^{3/2}$ A...

The above equation is Randles-Sevcik equation ...

where.

∴ i_p = Peak current.

∴ n = no. of electron in ...

∴ A = Area of electro...

∴ C = concentration o...

∴ D = Diffusion coeffic... electroactive sp...

∴ V = Scan rate.

$$E_{1/2} = \frac{E_{pa} + E_{pc}}{2}.$$

→ Qualitative analysis.